LINMA2415 Quantitative Energy Economics

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Academic year 2025-2026 - Q2

UCLouvain

Contents

Reminder about optimization

1.1 Formulations

Definition 1.1. The primal problem is the original optimization problem that we want to solve. It is formulated as:

$$\begin{aligned} \max_x \quad & c^T x \\ \text{s.t.} \quad & Ax \leq b \\ & x \geq 0 \end{aligned} \tag{1.1}$$

Definition 1.2. The dual problem is derived from the primal problem. It is formulated as:

$$\begin{aligned} \min_y \quad & b^T y \\ \text{s.t.} \quad & A^T y \geq c \\ & y \geq 0 \end{aligned} \tag{1.2}$$

Another way to look at the dual problem is by using the Lagrangian function. Consider now the primal under the form:

$$\begin{aligned} p^* = \min \quad & f_0(x) \\ \text{s.t.} \quad & f_i(x) \leq 0, \quad i = 1, \ldots, m \\ & h_j(x) = 0, \quad j = 1, \ldots, p \end{aligned} \tag{1.3}$$

Definition 1.3. The Lagrangian function is defined as:

$$L(x, \lambda, \nu) = f_0(x) + \sum_{i=1}^{m} \lambda_i f_i(x) + \sum_{j=1}^{p} \nu_j h_j(x) \tag{1.4}$$

Definition 1.4. The Lagrange dual function is defined as:

$$g(\lambda, \nu) = \inf_x L(x, \lambda, \nu) \tag{1.5}$$

Definition 1.5. The Lagrange dual problem is defined as:

$$\begin{aligned} d^* = \max \quad & g(\lambda, \nu) \\ \text{s.t.} \quad & \lambda \geq 0 \end{aligned} \tag{1.6}$$

1.2 Optimality conditions

Definition 1.6. Weak duality is when $d^* \leq p^*$, it always holds and can be used to find bounds on the optimal value of the primal problem.

Definition 1.7. Strong duality is when $d^* = p^*$, does not holds in general, but usually holds for convex problems.

Definition 1.8. Complementary slackness is a condition that holds at optimality, it states that for each constraint, either the constraint is active (i.e. $f_i(x) = 0$) or the corresponding dual variable is zero (i.e. $\lambda_i = 0$).

$$0 \leq \lambda_i^* \perp f_i(x^*) \leq 0 \tag{1.7}$$

Definition 1.9. The KKT conditions are a set of necessary conditions for optimality. Consider this problem:

$$\begin{aligned} \max\ & f(x,y) \\ \text{s.t. }\ & Ax + By \leq b \qquad (\lambda) \\ & Cx + Dy = d \qquad (\mu) \\ & x \geq 0 \qquad (\lambda_2) \end{aligned} \tag{1.8}$$

The KKT conditions are:

$$\begin{aligned} \mu \perp & Cx + Dy - d = 0 \\ 0 \leq \lambda \perp & Ax + By - b \leq 0 \\ 0 \leq x \perp & \lambda^T A + \mu^T C - \nabla_x f(x,y)^T \geq 0 \\ y \perp & \lambda^T B + \mu^T D - \nabla_y f(x,y)^T = 0 \end{aligned} \tag{1.9}$$

And the Lagrangian function can be written as:

$$L(x, y, \lambda, \mu) = f(x,y) + \lambda^T(b - Ax - By) + \mu^T(d - Cx - Dy) + \lambda_2^T x \tag{1.10}$$

Power systems

In our world, everything works with electricity and to be able to use it, there is complex systems behind it. In this chapter, we will see how the electricity is produced, transmitted and consumed. Usually a power system work like this: Electricity is produced in power plants, then it is transmitted through high voltage lines to substations, where it is transformed to lower voltage and distributed to consumers. The electricity is consumed by households, industries and other consumers. To ensure that the electricity supply meets the demand, there are reserves that can be activated in case of a sudden increase in demand or a decrease in supply. All of this electricity is exchanged throught financial markets, where producers and consumers can buy and sell electricity and other services.

2.1 Production

Electricity can be produced in different ways, there is two main categories of production: renewable and non-renewable. Renewable energy sources are those where the "fuel" is naturally present and is conceptually infinite, such as solar, wind, hydro, geothermal and biomass. Non-renewable energy sources are those where the "fuel" is extracted from the earth and is finite, such as coal, oil and natural gas. Nuclear energy is also considered non-renewable, although it does not emit greenhouse gases. Each type of production has its own characteristics and costs, and they are all important for the functioning of the power system. We can measure electricity by two main quantities: power and energy.

Definition 2.1. Power (MW) is the rate at which electricity is produced or consumed at a given moment. It is the instantaneous amount of electricity being generated or used.

Definition 2.2. Energy (MWh) is the total amount of electricity produced or consumed over a period of time. It is the integral of power over time.

Every type of production has its own structure and thus obey to different constraints to produce electricity. For example, here are some of the constraints that we can find in different types of production:

- **Fossil fuel power plants**:
  - Min/max power output
  - Ramp constraints (how fast the power can be increased or decreased)
  - Min up/down time (how long the plant must be on or off before it can change state)

- **Storage**:
  - Max charge/discharge power
  - Energy capacity (how much energy can be stored)
- **Renewables**:
  - Nameplate capacity (max power output under ideal conditions)
  - Capacity factor (actual power output as a fraction of nameplate capacity, depends on weather conditions)

But the production of electricity is not free. There are different types of costs associated with the production of electricity, such as fuel costs, fixed costs etc.

Definition 2.3. Variable cost (€/h) is the cost that depends on the amount of electricity produced. For example, fuel costs.

Definition 2.4. Fixed cost (€) is the cost that does not depend on the amount of electricity produced. For example, construction costs (€), overnight cost (€/kW (invested kW)).

Definition 2.5. Marginal cost (€/MWh) is the cost of producing one additional megawatt-hour of electricity. It is the derivative of the variable cost with respect to the power output.

Definition 2.6. Average cost (€/MWh) is the total cost per unit of electricity outputed.

Definition 2.7. The **merit order curve** is a graph that shows the marginal cost of production for different types of power plants. It is used to determine which power plants should be used to meet the demand at a given time, with the cheapest ones being used first.

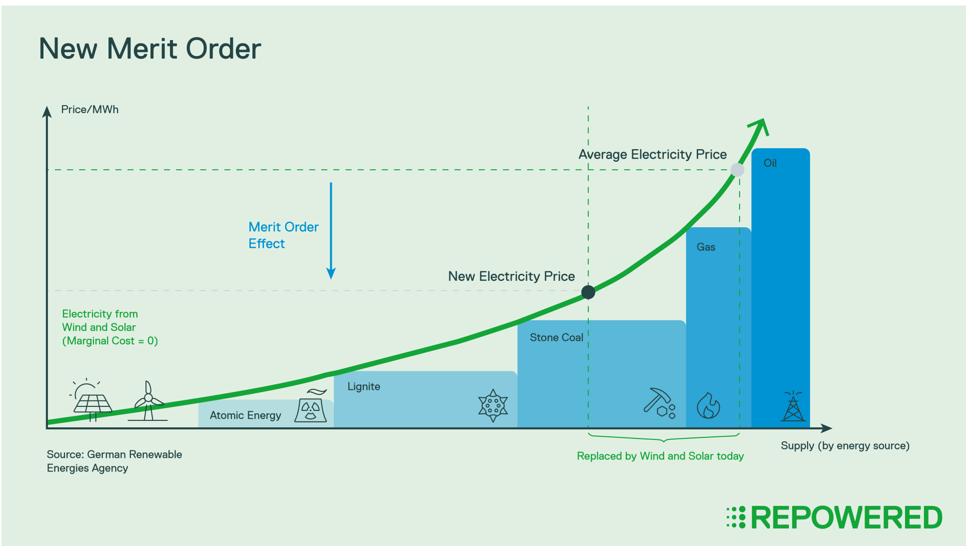

Figure 2.1: Example of merit order curve

2.2 Transmission and distribution

Definition 2.8. Transmission is the process of transporting electricity from the power plants to the substations. It is done through high voltage lines.

Definition 2.9. Distribution is the process of transporting electricity from the substations to the consumers. It is done through low voltage lines.

The transmission and distribution of electricity can be represented by a graph.

Definition 2.10. Buses are the nodes of the graphs, they represent the substations, producers and consumers. The power should be balanced at each bus, meaning that the power injected by the producers should be equal to the power consumed by the consumers and the power flowing through the lines connected to the bus.

Definition 2.11. Lines are the edges of the graph, they represent the transmission lines. They have a can have capacity.

The electricity flows through the lines according to the laws of physics, and the flow on each line is determined by the power injections at the buses and the impedance of the lines.

Definition 2.12. Power flow equations determine a mapping $f = P(r)$ of power injections r at the buses to power flows f on the lines.

Definition 2.13. DC power flow equations are an approximation of the power flow equations $f = P(r)$ by a linear mapping.

2.3 Consumption

Like the production of electricity, the consumption of electricity also has its own characteristics.

Definition 2.14. Consumer benefit (€/h) is the benefit that a consumer gets from consuming a certain amount of electricity.

Definition 2.15. Marginal benefit (€/MWh) is the benefit of consuming one additional megawatt-hour of electricity. It is the derivative of the consumer benefit with respect to the power consumption.

Definition 2.16. Inverse demand / marginal benefit function maps power consumption Q to marginal benefit $MB(Q)$.

Definition 2.17. Demand function maps prices of power v to quantity Q, $Q(v)$. It is the inverse mapping of the inverse demand function.

Definition 2.18. Elasticity is the sensitivity of the demand $Q(v)$ to changes in price v. It is defined as:

$$\epsilon = \frac{\frac{dQ}{dv}}{\frac{Q}{v}} \tag{2.1}$$

Definition 2.19. Average Value of Lost Load is the average cost to consumers of not having electricity. It is used to estimate the cost of power outages and to determine the value of reliability in the power system.

Definition 2.20. Value of Lost Load (VOLL) is the cost to consumers of not having electricity for a specific amount of time.

2.4 Reserves

With the rise of renewable energy sources, the uncertainty in the power system has increased, and the need for reserves has become more important. **Contingencies** (failures of components in the power system) can cause sudden changes in the supply or demand of electricity, and reserves are needed to ensure that the power system can still operate reliably. There are different types of reserves, that can be activated in different time frames:

- **Primary reserves** are activated and operationnal within seconds to few minutes after a contingency, they are used to stabilize the frequency of the power system. It is the first line of defense. It is usually provided by batteries for example because they are not limited by ramp constraints and can respond very quickly.

- **Secondary reserves** are activated within seconds but are only fully available within 5-10 minutes after a contingency. They can be supplied by spining (online generators) or non-spining reserves (offline generators but can be brought online quickly).

- **Tertiary reserves** are available within 15 minutes.

These reserves are activated sequentially. And on the following graph, we can see how the different reserves have been activated over a day in the Belgian power system.

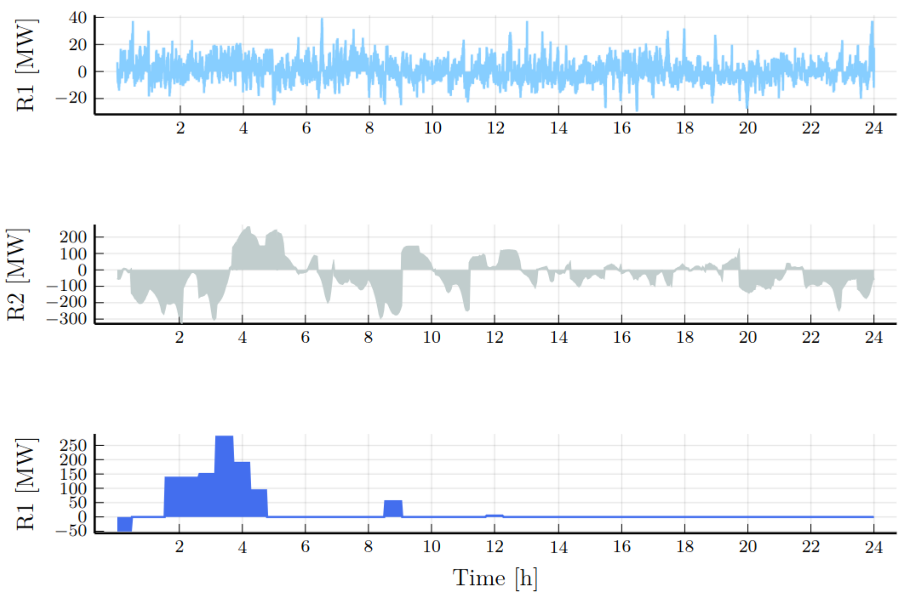

Figure 2.2: Example of activation of reserves in the Belgian power system

2.5 Multi-stage operations

The financial markets for electricity are organized in different time frames with different products exchanged at each time frame. This can be resumed in the following figure:

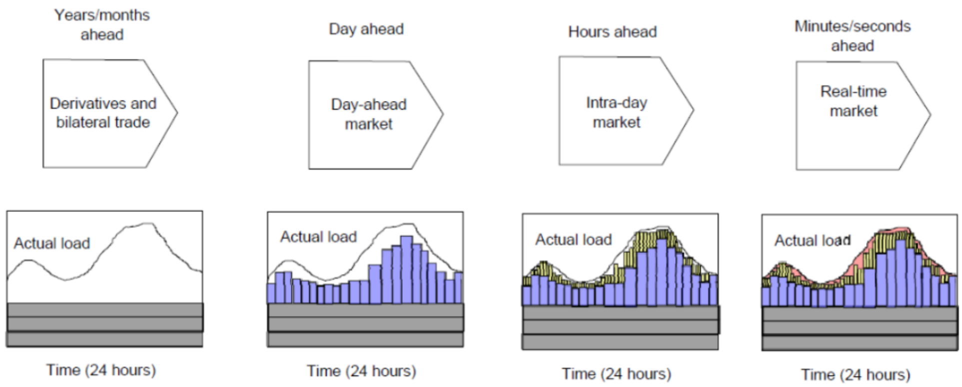

Figure 2.3: Order of the markets for electricity

Economic Dispatch

The economic dispatch problem is the problem of determining the optimal production of electricity by the different power plants to meet the demand at the lowest cost, while respecting the constraints of the power system. It is the simplest model that models a power system.

3.1 Formulation

The economic dispatch problem aims to maximize the total welfare of the system, which is the total benefit of the consumers minus the total cost of the producers. It can be formulated as follows:

$$
\begin{aligned}
\max \quad & \sum_{l \in L} \int_0^{d_l} MB_l(x)dx - \sum_{g \in G} \int_0^{p_g} MC_g(x)dx \\
\text{s.t.} \quad & \sum_{l \in L} d_l - \sum_{g \in G} p_g = 0 \quad [\lambda] \\
& 0 \le d_l \le D_l, \quad \forall l \in L \quad [\nu_l] \\
& 0 \le p_g \le P_g, \quad \forall g \in G \quad [\mu_g]
\end{aligned}
\tag{3.1}
$$

Where L is the set of loads, G is the set of generators. MC_g is the increasing marginal cost and MB_l is the decreasing marginal benefit. The KKT conditions for this problem are:

$$
\begin{aligned}
0 \le p_g \perp MC_g(p_g) - \lambda + \mu_g \ge 0 \\
0 \le \mu_g \perp P_g - p_g \ge 0 \\
0 \le d_l \perp -MB_l(d_l) + \lambda + \nu_l \ge 0 \\
0 \le \nu_l \perp D_l - d_l \ge 0 \\
\lambda \perp \sum_{g \in G} p_g - \sum_{l \in L} d_l = 0
\end{aligned}
\tag{3.2}
$$

There exists a unique price λ (except in some cases) that clears the market, meaning that the supply equals the demand. This price is called the **market clearing price** and it is the price at which the total welfare is maximized. We can find this price by solving the economic dispatch problem or by using the KKT conditions. So there exists a threshold λ such that:

- If $0 < p_g < P_g$, then $MC_g(p_g) = \lambda$.
- If $0 < d_l < D_l$, then $MB_l(d_l) = \lambda$.

- If $p_g = 0$, then $MC_g(p_g) \geq \lambda$.
- If $d_l = 0$, then $MB_l(d_l) \leq \lambda$.
- If $p_g = P_g$, then $MC_g(p_g) \leq \lambda$.
- If $d_l = D_l$, then $MB_l(d_l) \geq \lambda$.

3.2 Competitive market equilibrium

Definition 3.1. A market is **perfectly competitive** if

- Agent are price takers, meaning that they cannot influence the market clearing price λ.
- Variable costs are convex and benefits are concave.
- Agent have access to public information.

Definition 3.2. The **aggregate cost** is the cheapest way to produce Q MW of power among a set of producers. It is defined as:

$$\begin{aligned} TC_G(Q) = \min & \sum_{g \in G} \int_0^{p_g} MC_g(x)dx \\ \text{s.t.} \quad & \sum_{g \in G} p_g = Q \\ & p_g \in dom(MC_g) \end{aligned} \tag{3.3}$$

And thus the **aggregate marginal cost** is $MC_G(Q) = TC'_G(Q)$

Definition 3.3. The **aggregate benefit** is the cheapest way to consume Q MW of power among a set of consumers. It is defined as:

$$\begin{aligned} TB_L(Q) = \max & \sum_{l \in L} \int_0^{d_l} MB_l(x)dx \\ \text{s.t.} \quad & \sum_{l \in L} d_l = Q \\ & d_l \in dom(MB_l) \end{aligned} \tag{3.4}$$

And thus the **aggregate marginal benefit** is $MB_L(Q) = TB'_L(Q)$. In a competitive market, the market clearing price λ is decided with the **market clearing condition**:

$$\lambda \perp \sum_{g \in G} p_g - \sum_{l \in L} d_l = 0 \tag{3.5}$$

Definition 3.4. A market is at **equilibrium** if

- No profitable opportunities for trade exist, meaning that no agent can increase their welfare by changing their production or consumption.
- The market clearing price is the price of a market in equilibrium.

- Agent have access to public information.

In a competitive market, the equilibrium is also called the **competitive equilibrium** and the price is the **competitive price**.

Definition 3.5. The **Producer surplus** is the difference between the revenue of a producer and its variable cost. It is defined as:

$$PS_G = \lambda q_G(\lambda) - \int_0^{q_G(\lambda)} MC_G(x)dx \tag{3.6}$$

Definition 3.6. The **Consumer surplus** is the difference between the benefit of a consumer and the price they pay. It is defined as:

$$CS_L = \int_0^{q_L(\lambda)} MB_L(x)dx - \lambda q_L(\lambda) \tag{3.7}$$

Power markets

4.1 Exchange VS Pool market

Financial markets for electricity are a central part of the power system, they allow producers and consumers to buy and sell electricity and other services. There is 3 different ways to trade electricity:

- **Bilateral trade** are agreements between a producer and a consumer to buy/sell a certain amount of electricity at a certain price and time.
- **Exchange** is a centralized market where producers and consumers can buy and sell electricity by submitting simple bids with simples rules to auctions.
- **Pool** is a centralized market where producers and consumers can buy and sell electricity by submitting multi-part bids with complex rules to auctions.

Bilateral trade is the simplest way to trade electricity but it is the least centralised because all the are done in peer to peer. It is not efficient and thus cannot be used in real time. Exchange and pool are more centralised and thus more efficient.

4.1.1 Exchange Market

In an exchange market the bids look like this: (p_b, q_b) it is a pair of price and quantity. Positive values indicate a buy bid and negative values indicate a sell bid. The market operator collects all the bids then solves the following problem to determine the market clearing price and quantity:

$$\begin{aligned}
\max_{\delta_b} \quad & \sum_{b \in B} p_b |q_b| \delta_b \\
\text{s.t.} \quad & \sum_{b \in B} q_b \delta_b = 0 \\
& 0 \leq \delta_b \leq 1 \qquad \forall b \in B
\end{aligned} \tag{4.1}$$

4.1.2 Pool market

In a pool market, the bids are more complex. For example, it can contains start-up costs, marginal costs and capacity. This would generate a triplet of information for each bid: (SC_g, MC_g, P_g), in that case the market operator would solve the following problem to determine the market clearing price and quantity:

$$\begin{aligned}
\max_{d_l, p_g, u_g} \quad & \sum_{l \in L} MB_l d_l - \sum_{g \in G} (SC_g u_g + MC_g p_g) \\
\text{s.t.} \quad & \sum_{l \in L} d_l - \sum_{g \in G} p_g = 0 \\
& 0 \leq d_l \leq D_l, \quad \forall l \in L \\
& 0 \leq p_g \leq P_g u_g, \quad \forall g \in G \\
& u_g \in \{0, 1\}, \quad \forall g \in G
\end{aligned} \tag{4.2}$$

4.2 Electricity auctions

In the electricity market, there exists different types of auctions.

Definition 4.1. The **first-price auction** is an auction where the price for each winning bid is the price of the bid itself.

Definition 4.2. The **second-price auction** is an auction where the price will be decided by the highest rejected bid.

Definition 4.3. The **uniform price auction** is an auction where an uniform price is decided λ, and every winning bid will pay or be paid at λ €/MWh. It is used in exchange markets.

Both the uniform price auction and the pay as bid auction are criticized for different reasons. Uniform pricing is criticized for example for having high price volatility. Where pay-as-bid is criticized for being discriminatory (different price for the same product) and being less efficient.

4.2.1 Blueprint of an electricity market

An electricity market is organized like this:

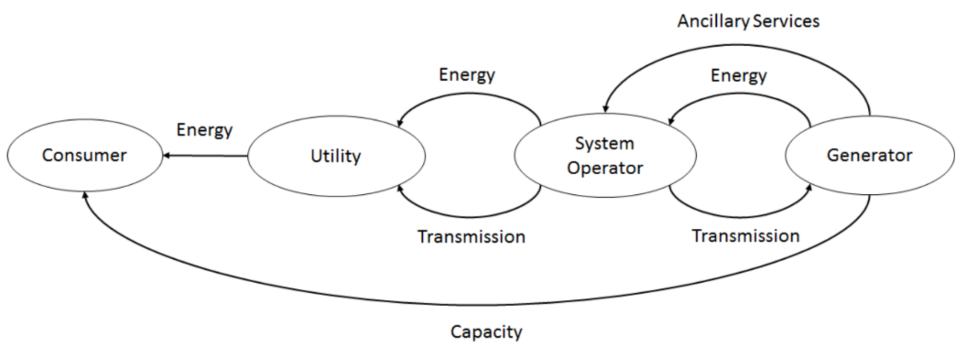

Figure 4.1: Blueprint of an electricity market

4.3 Nodal VS Zonal pricing

Until now, we have only considered market clearing problems without considering the physical configuration of the power system. In reality, the power system is a complex network of buses and lines, and the flow of electricity is determined by the laws of physics. This can lead to congestion in the lines. To manage this congestion and establish fair pricing, there are two main approaches: nodal pricing and zonal pricing.

4.3.1 Nodal pricing

Nodal pricing, is as the name suggests, a pricing mechanism where the price of electricity is determined at each node (or bus) in the power system. The price at each node reflects the marginal cost of supplying electricity to that node, this takes into account the cost of production, the cost of transmission and the congestion in the lines.

4.3.2 Zonal pricing

Zonal pricing, on the other hand, groups multiple nodes into zones and determines a single price for each zone. This approach is motivated by the need to simplify the pricing process and reduce the number of markets to be managed. This approach simplifies the pricing process but may not capture the full complexity of the power system and usually needs to be complemented with other mechanisms to manage congestion, such as financial transmission rights or redispatch.

Power flow

5.1 Network modeling

The power system can be modeled as a graph, where the nodes represent the buses and the edges represent the transmission lines. This graph can be characterized by some matrices. The one that interests us is the **Laplacian matrix** L, but to define it we need to define the **incidence matrix** A and the **degree matrix** D first. Let's consider a graph $G = (N, E)$, where N is the set of nodes and E is the set of edges.

Definition 5.1. The **incidence matrix** A is a matrix that describes the connectivity of the graph. It has one row and one column for each node. Each element of the matrix is defined as follows:

$$A_{ij} = \begin{cases} 1 & \text{if}(i, j) \in E \\ 0 & \text{otherwise} \end{cases} \tag{5.1}$$

Definition 5.2. The **degree matrix** D is a diagonal matrix that contains the degree of each node. The degree of a node is the number of edges connected to it. It is defined as:

$$D_{ii} = \sum_j A_{ij} \tag{5.2}$$

Definition 5.3. The **Laplacian matrix** L is defined as the difference between the degree matrix and the incidence matrix:

$$L = D - A \tag{5.3}$$

It is a positive semi-definite matrix that captures the connectivity of the graph.

Theorem 5.4. The multiplicity of the eigenvalue $\lambda = 0$ in the Laplacian of a graph is equal to the number of connected components of the graph

5.2 Circuits

The state of an electrical circuit can be described by the voltage at each node and the current flowing through each edge. The edge have a passive role as electricity flows through them. And the nodes have an active role as they inject or consume electricity. To know the state of a circuit, we need to know the voltage at each node compared to a reference node (usually called the ground) and the current flowing through each edge.

5.3 The power flow equations

The power flow equations are based on the laws of physics, on the Kirchhoff's law.

Definition 5.5. The **Kirchhoff's current law** states that the total current entering a node must be equal to the total current leaving the node. Mathematically, it can be expressed as:

$$\sum I = 0 \tag{5.4}$$

Definition 5.6. The **Kirchhoff's voltage law** states that the total voltage around any closed loop in a circuit must be equal to zero. Mathematically, it can be expressed as:

$$\sum V = 0 \tag{5.5}$$

5.4 DC power flow using bus angles

AC systems are caracterized by the fact that voltage is a sinusoidal wave, at every bus in the system, this wave can be different. A sinusoidal wave is caracterized by two things at each bus m:

- a magnitude: $|V_m|$
- a phase angle: θ_n

For this model, we are interrested in the phase angles. We follow the intuition that the power flow on a line is proportional to the difference of phase angles between the two ends of the line (i.e. $P_{mn} \propto \theta_m - \theta_n$). This equation tells us that when the phase angle at bus m is higher than the phase angle at bus n, then power will flow from bus m to bus n.
To caracterize the power flow on each line, we have the AC power flow equations:

$$P_m = |V_m| \sum_n |V_n| (G_{mn} \cos(\theta_m - \theta_n) + B_{mn} \sin(\theta_m - \theta_n)) \tag{5.6}$$

$$Q_m = |V_m| \sum_n |V_n| (G_{mn} \sin(\theta_m - \theta_n) - B_{mn} \cos(\theta_m - \theta_n)) \tag{5.7}$$

where G_{mn} and B_{mn} are the conductance and susceptance of the line between bus m and bus n. These equations are non-linear and non-convex, which makes them difficult to add in a optimization problem. To simplify them, we can make some assumptions:

- resistances are negligible: $G_{mn} \approx 0$
- angle differences are small: $\sin(\theta_m - \theta_n) \approx \theta_m - \theta_n$ and $\cos(\theta_m - \theta_n) \approx 1$
- voltage magnitudes are constant: $|V_m| \approx |V_n| \approx 1$

Theses assumptions lead us to the DC power flow equations:

$$P_m = \sum_n B_{mn} (\theta_m - \theta_n) \tag{5.8}$$

We can also write this equation in terms of reactance $X_{mn} = \frac{1}{B_{mn}}$:

$$P_m = \sum_{n \neq m} \frac{1}{X_{mn}}(\theta_m - \theta_n) \tag{5.9}$$

And thus we can write the power flow on each line $k = (m, n)$ as:

$$P_{mn} = \frac{1}{X_{mn}}(\theta_m - \theta_n) = B_{mn}(\theta_m - \theta_n) \tag{5.10}$$

When we use these in an optimization problem, we need to define a reference bus, which is a bus where we set the phase angle to zero. This is because the power flow equations are invariant to a shift in all the phase angles, so we need to fix one of them to have a unique solution.
We can also write the DC power flow equations in matrix form as follows:

$$P = T\theta \tag{5.11}$$

Where P is the vector of power injections at each bus, θ is the vector of phase angles at each bus and T is the weighted Laplacian matrix.

5.5 DC power flow using lines flows

The previous model with bus angles is criticized because angles are artificial variables and are not directly observable in markets. Plus usually we want to constraint the flow on the lines by these inequalities: $-\bar{F}_k \leq P_k \leq \bar{F}_k$. We want to rewrite $P = T\theta$ in terms of line flows, so we write $P_L = MT^{-1}P$ where P_L is the vector of line flows and MT^{-1} is the PTDF matrix and P is the vector of power injections.
Intuitively, the PTDF matrix represents mathematicaly by this sentence "If I inject $1MW$ at bus n and remove $1MW$ at the reference bus, how much flow appears on line k?". The PTDF of bus n on line k denoted F_{kn} is defined as follows :

$$F_{kn} = M_k(T_n^{-1}) \tag{5.12}$$

Where M_k is the row k of the matrix M and T_n^{-1} is the column n of the matrix T^{-1}. The matrix T is the weighted Laplacian matrix and the matrix M is defined as follows:

$$M_{kn} = \begin{cases} \frac{1}{X_k} & \text{if } k = (n, \dot{}), n \neq h \\ -\frac{1}{X_k} & \text{if } k = (\dot{}, n), n \neq h \\ 0 & \text{otherwise} \end{cases} \tag{5.13}$$

Where h is the reference bus. MAYBE CAN BE IMPROVED

5.6 DC optimal power flow

Until now we have only considered optimization problems without considering the physical configuration of the power system, like in the economic dispatch problem. The DCOPF (DC Optimal Power Flow) problem, is an optimization problem that maximizes the total welfare like in the economic dispatch problem, but it also takes into account the physical configuration of the power system by including a linearization of the kirchhoff's laws and the transmission limits constraints. To modelize that, there is two equivalent model. One based on the PTDF (Power Transfer Distribution Factor) and one based on the bus angles. Let us define how the transmission system is represented as a directed graph:

- N: set of nodes (buses)
- K: set of edges (lines) denoted $k = (n_1, n_2)$
- G_n: set of generators located at node n, $G = \cup_{n \in N} G_n$
- L_n: set of loads located at node n, $L = \cup_{n \in N} L_n$

The decision variables of the DCOPF problem will be p_g the amount of power produced by generator g, d_l the amount of power consumed by load l.

5.6.1 Power Transfer Distribution Factor model

For the PTDF model, we will interrest us to the nodal injections in the model. For that, we will choose a "hub node", which is a reference node that will absorb all the injections. The nodal injection at node n is defined as:

$$r_n = \sum_{g \in G_n} p_g - \sum_{l \in L_n} d_l \tag{5.14}$$

To follow the kirchhoff's laws, we need to add the following constraints:

$$\sum_{n \in N} r_n = 0 \tag{5.15}$$

Then, based on the PTDF (F_{kn}) that represents the amount of power flowing on line k as a result of shipping unit of power from node n to hub node, we can determine the flow on each line k as follows:

$$f_k = \sum_{n \in N} F_{kn} r_n \tag{5.16}$$

And we also need to add the transmission limits constraints:

$$-T_k \leq f_k \leq T_k \tag{5.17}$$

The full DCOPF model formulation with the PTDF is as follows:

$$\begin{aligned}
\max \sum_{l \in L} \int_0^{d_l} & MB_l(x)dx - \sum_{g \in G} \int_0^{p_g} MC_g(x)dx \\
\text{s.t.} \quad & f_k \leq T_k \quad \forall k \in K && [\lambda_k^+] \\
& -f_k \leq T_k \quad \forall k \in K && [\lambda_k^-] \\
& f_k - \sum_{n \in N} F_{kn} r_n = 0 \quad \forall k \in K && [\psi_k] \\
& r_n - \sum_{g \in G_n} p_g + \sum_{l \in L_n} d_l = 0 \quad \forall n \in N && [\rho_n] \\
& \sum_{n \in N} r_n = 0 && [\phi] \\
& 0 \leq p_g \leq P_g \quad \forall g \in G && [\mu_g] \\
& 0 \leq d_l \leq D_l \quad \forall l \in L && [\nu_l]
\end{aligned} \tag{5.18}$$

The KKT conditions for this problem are:

$$
\begin{aligned}
0 \leq p_g \perp MC_g(p_g) - \rho_{n(g)} + \mu_g \geq 0 \\
0 \leq d_l \perp -MB_l(d_l) + \rho_{n(l)} + \nu_l \geq 0 \\
0 \leq \lambda_k^+ \perp T_k - f_k \geq 0 \\
0 \leq \lambda_k^- \perp T_k + f_k \geq 0 \\
0 \leq \mu_g \perp P_g - p_g \geq 0 \\
0 \leq \nu_l \perp D_l - d_l \geq 0 \\
f_k \perp f_k - \sum_{n \in N} F_{kn} r_n = 0 \\
\rho_n \perp r_n - \sum_{g \in G_n} p_g + \sum_{l \in L_n} d_l = 0 \\
\perp \lambda_k^- - \lambda_k^+ + \psi_k = 0 \\
\perp \phi + \rho_n - \sum_{k \in K} F_{kn} \psi_k = 0
\end{aligned}
\tag{5.19}
$$

With these, we can see that there exists a threshold ρ_n for all n such that:

- If $0 < p_g < P_g$, then $\rho_n = MC_g(p_g)$
- If $0 < d_l < D_l$, then $\rho_n = MB_l(d_l)$
- If $p_g = 0$, then $\rho_n \leq MC_g(p_g)$
- If $d_l = 0$, then $\rho_n \geq MB_l(d_l)$
- If $p_g = P_g$, then $\rho_n \geq MC_g(p_g)$
- If $d_l = D_l$, then $\rho_n \leq MB_l(d_l)$

We can interpret the dual variables as follows:

- ϕ is the dual variable associated with the global balance constraint. It acts as a reference price level since only nodal price differences are economically meaningful.
- λ_k^+ and λ_k^- are congestion shadow prices associated with the transmission limits. They measure the marginal welfare gain of increasing the capacity of line k.
- ρ_n represents the locational marginal price (LMP) at node n, i.e. the marginal value of injecting one additional MW at node n.

5.6.2 Bus angle model

For the bus angle model, also known as the reactance model, we will interrest us to the differences of phase angles in the network. For that, we will choose a "hub node", its choice won't affect the results. The phase angle at node n is denoted θ_n. The flow on line $k = (m, n)$ is defined as:

$$f_{(m,n)} = B_{(m,n)}(\theta_m - \theta_n) \tag{5.20}$$

Where $B_{(m,n)}$ is the susceptance of line (m, n) and $\theta_{hub} = 0$. Then, we need to define the conservation of energy constraints at each node:

$$\sum_{g \in G_n} p_g + \sum_{k=(.,n)} f_k = \sum_{k=(n,.)} f_k + \sum_{l \in L_n} d_l \tag{5.21}$$

The full DCOPF model formulation with the bus angle is as follows:

$$\begin{aligned}
\max \sum_{l \in L} \int_0^{d_l} MB_l(x)dx - \sum_{g \in G} \int_0^{p_g} MC_g(x)dx & \\
\text{s.t.} \quad f_k \le T_k \quad \forall k \in K & \quad [\lambda_k^+] \\
- f_k \le T_k \quad \forall k \in K & \quad [\lambda_k^-] \\
f_{(m,n)} - B_{(m,n)}(\theta_m - \theta_n) = 0 \quad \forall (m,n) \in K & \quad [\gamma_{(m,n)}] \\
\sum_{k=(n,.)} f_k + \sum_{l \in L_n} d_l - \sum_{g \in G_n} p_g - \sum_{k=(.,n)} f_k = 0 & \quad [\rho_n] \\
0 \le p_g \le P_g \quad \forall g \in G & \quad [\mu_g] \\
0 \le d_l \le D_l \quad \forall l \in L & \quad [\nu_l]
\end{aligned} \tag{5.22}$$

Pricing transmission

In the DCOPF problem, we have seen that there exists a different price ρ_n for each node n, this different price at each node is induced by the transmission constraints. Intuitively, if a node has cheapgeneration and cannot export much, its price will be low and if a node has high demand cannot import enought cheap electricity, its price will be high. With that we can introduce a new market operator.

Definition 6.1. The **Transmission market operator (TSO)** is the market operator that manages the grid. It can be seen as an agent that buys electricity where it is cheap and sells it where it is expensive while respecting the transmission constraints.

With that, we can rewrite the DCOPF problem as a decentralized problem with different market operators: producers, consumers, TSO and the auctionneer. Each of these agents will solve their own optimization problem and the market clearing condition will be the same as before. The TSO will solve the following problem:

$$\begin{aligned} \max \quad & \sum_{n \in N} -\rho_n r_n \\ \text{s.t.} \quad & f_k - \sum_{n \in N} F_{kn} r_n = 0 \quad \forall k \in K \\ & \sum_{n \in N} r_n = 0 \\ & -T_k \leq f_k \leq T_k \quad \forall k \in K \end{aligned} \tag{6.1}$$

The producers will solve the following problem:

$$\begin{aligned} \max \quad & p_g \rho_{n(g)} - \int_0^{p_g} MC_g(x) dx \\ \text{s.t.} \quad & 0 \leq p_g \leq P_g \quad \forall g \in G \end{aligned} \tag{6.2}$$

The consumers will solve the following problem:

$$\begin{aligned} \max \quad & \int_0^{d_l} MB_l(x) dx - d_l \, \rho_{n(l)} \\ \text{s.t.} \quad & 0 \leq d_l \leq D_l \quad \forall l \in L \end{aligned} \tag{6.3}$$

And the auctionneer will solve the following problem:

$$\max \quad \rho_n \left(r_n + \sum_{l \in L_n} d_l - \sum_{g \in G_n} p_g \right) \tag{6.4}$$

But solving all these problems requires a lot of communication between the different agents, and thus it is not very practical. Plus the TSO needs to take position in the different markets and has lots of opportunities to abuse of its position.

6.1 Nodal pricing

In nodal pricing also called local marginal pricing (LMP) , there is an uniform price auction then the TSO will solve the DCOPF that will decide the price ρ_n at each node n. If all the agents bid truthfully, then the solution outputed by the LMP, will be the same as the optimal solution of the DCOPF problem. In this configuration, if there is no congestion, prices will tend to be equals at all nodes, but if there is congestion, prices will differ between nodes.
The congestion on the lines and the choices made by the TSO to manage this congestion generates a cost for the consumers and a rent for the producers.

Definition 6.2. The **congestion rent** is the profit that the TSO makes by buying electricity where it is cheap and selling it where it is expensive. It is defined as:

$$\sum_{n\in N} \rho_n \left(\sum_{l\in L_n} d_l - \sum_{g\in G_n} p_g \right) \tag{6.5}$$

6.2 Zonal pricing

Nodal pricing is criticized for being too complex because it generates too many prices. Plus the multiplicity of small markets leads to a lack of liquidity in each market but also to a lot of opportunities for market power abuse. So an altenative to nodal pricing is zonal pricing, where the nodes are grouped into zones and a single price is determined for each zone. The zonal DCOPF problem is formulated as follows:

$$\begin{aligned}
\max & \sum_{l\in L} \int_0^{d_l} MB_l(x)dx - \sum_{g\in G} \int_0^{p_g} MC_g(x)dx \\
\text{s.t.} \quad & f_k \leq T_k \quad \forall k \in K && [\lambda_k^+] \\
& -f_k \leq T_k \quad \forall k \in K && [\lambda_k^-] \\
& f_k - \sum_{n\in N} F_{kn} r_n = 0 \quad \forall k \in K && [\psi_k] \\
& r_z - \sum_{g\in G_z} p_g + \sum_{l\in L_z} d_l = 0 \quad \forall z \in Z && [\rho_z] \\
& \sum_{n\in N} r_n = 0 && [\phi] \\
& 0 \leq p_g \leq P_g \quad \forall g \in G && [\mu_g] \\
& 0 \leq d_l \leq D_l \quad \forall l \in L && [\nu_l]
\end{aligned} \tag{6.6}$$

In zonal pricing, there is an uniform price auction then the TSO will solve the zonal DCOPF that will decide the price ρ_z at each zone z. Within each zone, Kirchhoff's laws are ignored this can cause congestion problems. These congestion problems needs to be solved by the TSO before the time of delivery by redispatching the production and consumption. The redispatch works with a pay as bid auction, where the market participants are forced to bid their true marginal cost/benefit. The redispatch is thus financially neutral to market participants. The TSO will cover the difference between payments received and made. The **perfect cost-based redispatch** minimize the redispatch cost for the TSO and given the dispatch obtained under the zonal pricing (p_g^*, d_l^*), it is formulated as follows:

$$
\begin{aligned}
RD = \max \sum_{l \in L} \int_0^{d_l^* + \Delta d_l} MB_l(x)dx - \sum_{g \in G} \int_0^{p_g^* + \Delta p_g} MC_g(x)dx \\
\text{s.t.} \quad & f_k \leq T_k \quad \forall k \in K \\
& -T_k \leq f_k \leq T_k \quad \forall k \in K \\
& P_g - p_g^* - \Delta p_g \quad \forall g \in G \\
& D_l - d_l^* - \Delta d_l \quad \forall l \in L \\
& r_n + \sum_{g \in G_n(n)} (p_g^* + \Delta p_g) - \sum_{l \in L_n(n)} (d_l^* + \Delta d_l) = 0 \quad \forall n \in N \\
& f_k - \sum_{n \in N} F_{kn} r_n = 0 \quad \forall k \in K \\
& \sum_{n \in N} r_n = 0
\end{aligned}
\tag{6.7}
$$

6.3 Other formulations

6.3.1 Zonal pricing with Availability Transfer Capacities (ATC)

The formulation of the zonal pricing with ATC is the same as the one of the zonal pricing, but with an additional constraint that limits the flow between zones. The full formulation is as follows:

$$
\begin{aligned}
\max \sum_{l \in L} \int_0^{d_l} MB_l(x)dx - \sum_{g \in G} \int_0^{p_g} MC_g(x)dx \\
\text{s.t.} \quad & \sum_{a=(z,.)} f_a + \sum_{l \in L_z} d_l = \sum_{a=(.,z)} f_a + \sum_{g \in G_z} p_g \quad \forall z \in Z \quad & [\rho_z] \\
& -ATC_a \leq f_a \leq ATC_a \quad \forall a \in A & [\lambda_a^+, \lambda_a^-] \\
& 0 \leq p_g \leq P_g \quad \forall g \in G & [\mu_g] \\
& 0 \leq d_l \leq D_l \quad \forall l \in L & [\nu_l]
\end{aligned}
\tag{6.8}
$$

Where A is the set of inter-zonal lines and ATC_a is the capacity of line a that between two zones.

6.3.2 Flow-based (FB) zonal pricing

Unlike the zonal pricing with ATC, the flow-based zonal use the PTDF to model the flow between zones and thus the flow between zones influences the flow between other zones. Using zonal PTDF, the constraints of the flow between zones are as follows:

$$
-T_k \leq \sum_{z \in Z} \tilde{F}_{kz} \big(\sum_{g \in G_z} p_g - \sum_{l \in L_z} d_l \big) \leq T_k \quad \forall k \in K \tag{6.9}
$$

We need to define the zonal PTDF $\tilde{F}$, it is defined as follows:

$$
\tilde{F}_{kz} = \sum_{n \in N} \frac{\tilde{r}_n}{\sum_{n \in N} \tilde{r}_n} F_{kn} \tag{6.10}
$$

So the contribution of a node to the zonal PTDF is proportional to its nodal injection to the total nodal injection of the zone.

Unit commitment and nonconvexities

Until now, we considered only basic convex problems, like in the economic dispatch. But in reality, real power plant are much more complex systems and their cost functions introduces non-convexities. These non-convexities are for examples start-up costs, no load costs, minimum up and down times, ramping constraints, minimum stable generation, etc. To address these non-convexities, we introduce the unit commitment problem.

7.1 The unit commitment problem

The unit commitment problem is an optimization problem that determines the optimal schedule of power plants to meet the demand while minimizing the total cost. For that, we need to introduce times dependant variables:

- $p_{g,t}$: power output of generator g at time t
- $u_{g,t}$: binary variable that indicates whether generator g is on or off at time t
- $v_{g,t}$: binary variable that indicates whether generator g is starting up at time t
- $w_{g,t}$: binary variable that indicates whether generator g is shutting down at time t

This model with non-convexities uses binary variables and thus is a mixed-integer linear program (MILP). The formulation of the unit commitment problem is as follows:

$$\begin{aligned}
\min \; & \sum_g \sum_t (MC_g p_{g,t} + K_g u_{g,t} + S_g v_{g,t}) \\
\text{s.t.} \quad & \sum_g p_{g,t} = D_t \quad \forall t \\
& P_g^{\min} u_{g,t} \leq p_{g,t} \leq P_g^{\max} u_{g,t} \quad \forall g, t \\
& u_{g,t} = u_{g,t-1} + v_{g,t} - w_{g,t} \quad \forall g, t \\
& u_{g,t}, v_{g,t}, w_{g,t} \in \{0, 1\} \quad \forall g, t
\end{aligned} \tag{7.1}$$

For additional ramp constraint, we can add the following constraints:

$$\begin{aligned}
p_{g,t} &\leq p_{g,t-1} + R_g^+ - v_{g,t}(R_g^+ - R_{SU}^+) \quad \forall t \\
p_{g,t} &\geq p_{g,t-1} - R_g^- + w_{g,t}(R_g^- - R_{SD}^-) \quad \forall t
\end{aligned} \tag{7.2}$$

where R_g^+ and R_g^- are the ramp up and down limits of generator g, R_{SU}^+ and R_{SD}^- are the ramp up and down limits during start-up and shut-down respectively.
For additional minimum up and down time constraints, we can add the following constraints:

$$\begin{aligned}
\sum_{i=t-UT_g+1}^{t} v_{g,i} \leq u_{g,t} \quad \forall t \geq UT_g \\
\sum_{i=t-DT_g+1}^{t} w_{g,i} \leq 1 - u_{g,t} \quad \forall t \geq DT_g
\end{aligned} \tag{7.3}$$

Due to the heaviness of the unit commitment problem, and its time dependancies, the model needs to be solved ahead of time and thus it is used for the day-ahead market. Usually, it is solved once for the whole next day.

7.2 Market equilibrium under non-convexities

Finding the market equilibrium under non-convexities is a complex problem. Ideally, we want to obtain the four fundamental properties of markets.

- **Efficiency**: the market equilibrium should maximize the total welfare of the market.
- **Individual rationality**: the agents have no incentive to deviate from the market equilibrium.
- **Cost recovery**: every agent has a non-negative profit
- **Revenue adequacy**: the revenue of the market operator is non-negative

The **Impossibility theorem** states that:

Theorem 7.1. Once you introduce the discrete nature of power production, it becomes mathematically impossible to find fundamental prices that respect the 4 properties.

There exists multiples relaxation of the unit commitment problem that can be used to solve a convexified version of the problem.

7.2.1 Marginal pricing

The marginal pricing is the simplest way to convexify the problem, it consists in solving the unit commitment problem and then fixing the unit commitment variables to their optimal value and then solving the resulting convex problem to find the marginal prices. But the solution will not be an equilibrium. Some agent could still have financial incentives to deviate from the solution. We can measure these financial incentives by the **lost opportunity cost** of an agent.

Definition 7.2. The **lost opportunity cost** of an agent is the difference between the maximum possible profit of the agent and it's profit as the market clears.

$$\begin{aligned}
LOC_g(\pi_t) = \max \quad & \sum_t \left((\pi_t - MC_g)p_{g,t} - K_g u_{g,t} - S_g v_{g,t}\right) \\
& - \left(\sum_t \left((\pi_t - MC_g)p_{g,t}^* - K_g u_{g,t}^* - S_g v_{g,t}^*\right)\right)
\end{aligned} \tag{7.4}$$

with $(p^*_{g,t}, u^*_{g,t}, v^*_{g,t})$ the optimal solution of the unit commitment problem and π_t the decided price at time t.

7.2.2 Convex hull pricing

Convex hull pricing is a more complex way to find a market equilibrium under non-convexities. It consists in computing the optimal Lagrange multipliers of the market clearing constraints in the nonconvex problem.

Theorem 7.3. The Hogan theorem states that the convex hull prices are the uniform prices that minimize the total lost opportunity costs.

The solution of the convex hull pricing will lead to a dispatch that is the same as the one of the unit commitment problem, but with different prices. Like the marginal pricing, the dispatch obtained are welfare maximizing.

7.2.3 EUPHEMIA pricing

EUPHEMIA pricing is the pricing rule used in European day-ahead electricity markets. Contrary to marginal pricing or convex hull pricing with uplift payments, the European market design does not compensate participants with side payments. The objective is therefore not only to clear the market, but also to obtain a uniform market price that does not require make-whole payments.
The main idea is to avoid paradoxically accepted bids. A paradoxically accepted bid (PAB) is a bid that is accepted even though, at the market price, it would make a loss. Such a bid would require a make-whole payment to recover its costs. EUPHEMIA prevents this situation by adding constraints ensuring that non-convex bids cannot be accepted if they are out-of-the-money.
However, EUPHEMIA allows paradoxically rejected bids (PRB). A paradoxically rejected bid is a bid that is rejected even though it would have been profitable at the final market price. These rejected bids are not compensated for their opportunity cost. This is the main difference with convex hull pricing, where the objective is to reduce lost opportunity costs through uplift payments.
Thus, EUPHEMIA pricing can be understood as a form of integer programming pricing with an additional no-PAB constraint. Marginal bids still determine the uniform price, but non-convex bids that would need a make-whole payment are rejected. This ensures cost recovery and revenue adequacy, but it can reduce efficiency because some welfare-improving bids may be excluded from the final dispatch.

7.2.4 Comparison between models

In the start of the section, we have seen that it is impossible to find a market equilibrium that respect the 4 fundamental properties of markets. But they can respect some of them. Let's compare the different models:

Model	MP	MP with uplifts	CHP with uplifts	EUPHEMIA pricing
Efficiency	Yes	Yes	Yes	No
Individual Rationality	No	Yes	Yes	No
Cost Recovery	No	Yes	Yes	Yes
Revenue Adequacy	Yes	No	No	Yes

Cost recovery is very important for incentivizing market participation, which is why uplifts are used.

Ancillary services

Ancillary services are necessary services to support the transmission of electric power from seller to purchaser given the obligations of control areas to maintain reliable operations. There is multiples types of ancillary services such as reserves, scheduling and dispatch, voltage control, energy imbalance, etc.

8.1 Categories of reserves

With the rise of renewable energy, the uncertainty in the power system has increased, which has led to an increase in the need for reserves. Reserves are backup resources that can be called upon to maintain the balance between supply and demand in the power system. The reserves are there to ensure the reliability of the power system in the discrete case of unexpected events such as a generator failure or in the continuous case of forecast errors. There are 3 types of reserves: primary, secondary and tertiary reserves.

8.1.1 Primary reserves

Primary reserves, also known as frequency containment reserves, are the first line of defense against frequency deviations in the power system. It should be activated within maximum 30 seconds after the occurrence of a frequency deviation. It can be provided by generators that can quickly adjust their output by changing their rotor inertia, the effect is immediate. It can also be provided by frequency responsive automatic controllers that can adjust parameters in response to frequency deviations, the reaction is also immediate but can take a few seconds to be fully effective. Or even by automatic generation control (AGC) like load frequency control, they are updated every few seconds up to a minute. The primary reserves are activated automatically and are not controlled by the TSO, they are activated by the frequency deviation itself.

8.1.2 Secondary reserves

Secondary reserves, also known as frequency restoration reserves, are the second line of defense. They are classified into two categories: spinning and non-spinning reserves. Spinning reserves are the generators that are already online and can increase their output. And the non-spinning reserves are the generators that are offline but can be started up quickly. The secondary reserves are activated by the TSO and they are activated within 5 minutes after the occurrence of a frequency deviation. The secondary reserves are activated to restore the frequency to its nominal value and to free up the primary reserves.

8.1.3 Tertiary reserves

Tertiary reserves, are the third line of defense. They are notably composed of manual restoration services. Their maximum activation time is 15 minutes after the occurrence of a frequency deviation.

8.1.4 Sequential activation of reserves

When a frequency deviation occurs, the primary reserves are activated first, then the secondary reserves and finally the tertiary reserves. Each of them has constraints on the minimal activation time, the amount of power that can be provided and the duration of activation. Here is a schematic representation of the sequential activation of reserves:

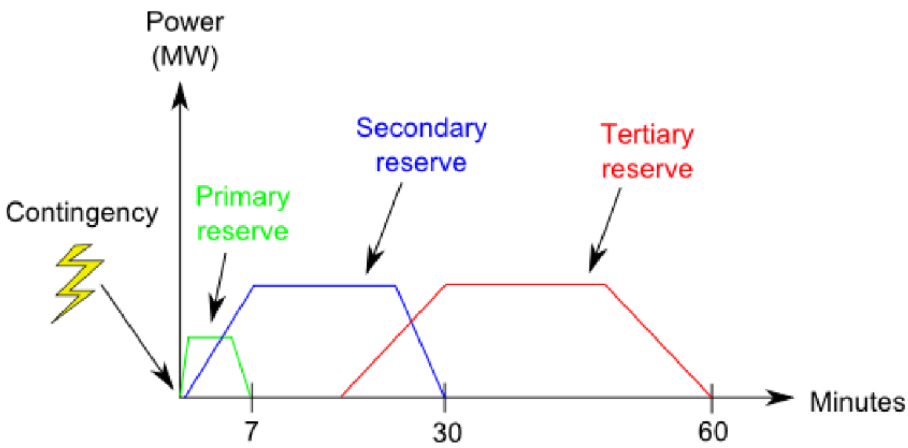

Figure 8.1: Sequential activation of reserves

8.1.5 Optimization models for reserve

It is possible to add the reserve decision to the existing economic dispatch models established in the previous chapters. For that generators can do what is called **curtailment**, which is to produce less than their maximum capacity in order to have some reserve capacity that can be activated if needed. This can be modeled by adding a new decision variable r_g that represents the amount of reserve provided by generator g. The optimization problem will then include the following constraints:

$$
\begin{aligned}
& p_g + r_g \leq P_g \quad \forall g \in G \\
& r_g \leq R_g \quad \forall g \in G \\
& \sum_{g \in G} r_g \geq R \\
& r_g \geq 0 \quad \forall g \in G
\end{aligned} \tag{8.1}
$$

Where R_g is the maximum reserve that generator g can provide in the allowed time of the reserve type (primary, secondary or tertiary) according to its ramp constraints.

R is the total reserve requirement that needs to be met. But curtailement is not the only way to provide reserve, it can also be provided in the other way. Curtailment offer upward reserve to cover a deficiency in production, but there is also downward reserve that can be provided by decreasing the production of a generator because there is too much production in the system. To show the differences between the two types of reserve, we will add an indices $+$ and $-$ to the reserve variable to represent upward and downward reserve respectively. We can also add another indices to represent the type of reserve (primary, secondary or tertiary). So we will have complex variables such as $r^+_{g,1}$ and $r^-_{g,1}$ that represent respectively the upward and downward reserve provided by generator g for the first type of reserve. This lead us to the economic dispatch with reserve problem defined like this:

$$
\begin{aligned}
EDR = \max & \sum_{l \in L} \int_0^{d_l} MB_l(x)dx - \sum_{g \in G} \int_0^{p_g} MC_g(x)dx \\
\text{s.t.} \quad & \sum_{l \in L} d_l - \sum_{g \in G} p_g = 0 && [\lambda] \\
& R_i^+ \leq \sum_{g \in G} r^+_{g,i} \quad \forall i \in \{1,2,3\} && [\mu_i^+] \\
& R_i^- \leq \sum_{g \in G} r^-_{g,i} \quad \forall i \in \{1,2,3\} && [\mu_i^-] \\
& r^+_{g,i}, r^-_{g,i} \leq R_g^i \quad \forall g \in G, i \in \{1,2,3\} \\
& p_g + \sum_{i \in \{1,2,3\}} r^+_{g,i} \leq P_g \quad \forall g \in G \\
& p_g - \sum_{i \in \{1,2,3\}} r^-_{g,i} \geq 0 \quad \forall g \in G
\end{aligned}
\tag{8.2}
$$

8.2 Market auction with reserve

8.2.1 Simultaneous auction for energy and reserve

Now that we added the reserve decision to the optimization problem, there is two market clearing constraints for the EDR. Like before, we have the market clearing constraint for energy $\sum_{l \in L} d_l - \sum_{g \in G} p_g = 0$. And now we add the market clearing constraint for reserve $\sum_{g \in G} r_g \geq R$. The simultaneous auction for energy and reserve is a market design where the energy and reserve markets are cleared at the same time.
In the exchange version of this new market, the producers and consumers will submit price and quantity bids for both energy and reserve while respecting their ramp constraints.
While in the pool version, the producers will submit ramp rates and costs, and the consumers will submit their demand and willingness to pay. After that, the TSO will solve the EDR problem and announce the market clearing price for power λ and the market clearing price for reserve μ.

8.2.2 Sequential clearing of energy and reserve

It is also possible to clear the energy and reserve markets sequentially. In this case, the reserve market is cleared first, then the energy market is cleared.

8.3 Balancing

Definition 8.1. Balancing is the task of adjusting power production and consumption in real-time to maintain the balance between supply and demand in the power system.

Balancing is closely related to reserve markets. Resources that committed reserve capacity in earlier markets can be activated in real time to compensate for forecast errors, renewable uncertainty, or unexpected outages. Two important actors participate in balancing markets:

- **Balancing Service Providers (BSPs)**: generators, storage units, or flexible consumers capable of adjusting their production or consumption in real time. BSPs submit balancing bids to the TSO.

- **Balancing Responsible Parties (BRPs)**: market participants responsible for keeping their portfolio balanced. If their real-time production or consumption deviates from their scheduled position, they must financially compensate the system.

The balancing market operates using **increment** and **decrement** bids. Which are increase in production or decrease in consumption for increment bids, and decrease in production or increase in consumption for decrement bids. The TSO activates these bids in real time to remove system imbalances.
This lead us to a new optimization problem, the balancing market model:

$$\begin{aligned} \max \; & \sum_{d \in D} MB_d \delta_d^- - \sum_{u \in U} MC_u \delta_u^+ \\ \text{s.t.} \; & \sum_{u \in U} \delta_u^+ - \sum_{d \in D} \delta_d^- = \Delta \qquad [\lambda] \\ & 0 \leq \delta_u^+ \leq \Delta_u \quad \forall u \in U \\ & 0 \leq \delta_d^- \leq \Delta_d \quad \forall d \in D \end{aligned} \tag{8.3}$$

Where Δ is the imbalance that needs to be covered, D is the set of decrement bids, U is the set of increment bids, δ_d^- is the amount of decrement bid d that is activated and δ_u^+ is the amount of increment bid u that is activated.
The market clearing price for balancing will be given by the dual variable λ associated with the market clearing constraint, it is called the **balancing price**. If the system faces an upward imbalance ($\Delta > 0$), the TSO activates increment bids in increasing order of marginal cost. Conversely, if the system faces a downward imbalance ($\Delta < 0$), decrement bids are activated.
An important observation is that balancing markets can also be interpreted as a real-time market clearing mechanism. Indeed, clearing increment and decrement bids is equivalent to re-clearing the supply and demand curves in real time after unexpected events occur.

Hedging risks

9.1 Forward contract

Definition 9.1. Forward contract are financial contracts that allows market participants to trade electricity at a predetermined price fixed in advance of the delivery. A forward contract is a bilateral agreement between two parties, it is characterized by a selling price f_t, a quantity x of traded electricity and a delivery time T (also called expiration date).

The advantages of forward contracts are that they allow market participants to hedge (se couvrir) against price volatility, they do not influence the real-time price of electricity and they can be traded. The different transactions that occur in a forward contract can be summarized in the following table:

Time	Producer	Consumer	Type of transaction	What happens
t	$+f_t x$	$-f_t x$	Financial	Contract emission
T	$-p_T x$	$+p_T x$	Financial	Reimbursement
T	$+p_T q$	$-p_T q$	Physical (TSO)	Real time costs
Total	$f_t x + p_T(q-x)$	$p_T(x-q) - f_t x$		

Suppose that the producer sells a contract for a quantity x at a price f_t at time t but produces q in real time for a price p_T. Then the producers would be paid $R = f_t x + p_T(q-x)$. So the producer should produce x to maximize its revenue, and thus hedges the risk of price volatility. The same applies for the consumer, it should consume x to minimize its cost and thus hedges the risk of price volatility.
Given risk neutral agents, the forward price f_t should be equal to the expected real-time price $\mathbb{E}[p_T|\xi_{[t]}]$, where $\xi_{[t]}$ is the state of the world at time t.

9.1.1 Contract for Differences

A contract for differences (CfD) is a special type of forward contract where the payment is only done at the delivery time T. In this type of contract, the buyer and the seller agree on a price f_t at time t but the payment is only done at time T. And only one transaction occurs at time T, the buyer pays the seller $(f_t - p_T)x$. This is an advantage if the buyer does not have the financial resources to pay at time t where the forward contract has the advantage of giving liquidity to the seller to produce the electricity. The different transactions that occur in a contract for differences can be summarized in the following table:

Time	Producer	Consumer	Type of transaction	What happens
t	0	0	Financial	Contract emission
T	$+(f_t - p_T)x$	$-(f_t - p_T)x$	Financial	Reimbursement
T	$+p_T q$	$-p_T q$	Physical (TSO)	Real time costs
Total	$f_t x + p_T(q - x)$	$p_T(x - q) - f_t x$		

9.2 Financial Transmission Rights (FTR)

Forward contracts are adequate for trading electricity at a fixed price in a market without congestion but in the case of congestion, additionnaly to the price of the forward contract, the market participants will need to pay the congestion rent. To hedge against this congestion rent, market participants can use financial transmission rights (FTR). Concretely, an FTR is a financial contract that gives the holder the right to ship power between two buses of the transmission network.
A FTR works a bit like a forward contract, the buyer and the seller agree on a price at time t to ship x units of power from bus A to bus B and the buyer. Then at time T, the seller pays the buyer the difference in price between bus A and bus B, so $(p_A - p_B)x$.
The FTR's are sell by the TSO and an allocation of FTR's must respect the transmission constraints. The FTR payment if feasible covers the congestion rent, so it is a good way to hedge against congestion rent.

9.3 Modelling risk aversion

Definition 9.2. A **risk measure** is a mapping $\xi : \Theta \to \mathbb{R}$ from a real-valued random variable to a real number that quantifies the risk of the random variable.

With this we can define some risk measures.

Definition 9.3. The **expected value** of a risk measure $\mathcal{R}$ is defined as:

$$\mathbb{E}[\mathcal{R}(\xi)] \tag{9.1}$$

It is the most commonly used risk measure.

Definition 9.4. The **worst-case payoff** of a risk measure $\mathcal{R}$ is defined as:

$$\max_{\omega \in \Omega} \xi(\omega) \tag{9.2}$$

Definition 9.5. The **value at risk** (VaR) is the greatest loss in portfolio value that can occur with probability α, it is defined as:

$$VaR_\alpha(\xi) = \min\{t | \mathbb{P}[\xi \leq t] \geq \alpha\} \tag{9.3}$$

Value at risk is highly sensitive with respect to market data. To overcome this problem and have a more stable risk measure, we can use the following risk measure.

Definition 9.6. The **Conditional Value at Risk** (CVaR) is the expectation of losses, conditional on losses being greater then VaR, it is defined as:

$$CVaR_\alpha(\xi) = \mathbb{E}_{P_\alpha}[\xi] \tag{9.4}$$

where

$$P_\alpha(\omega) = \begin{cases} \frac{\mathbb{P}[\xi \le t] - \alpha}{1-\alpha} & \text{if } \omega \ge VaR_\alpha(\xi) \\ 0 & \text{otherwise} \end{cases} \tag{9.5}$$

Definition 9.7. We say that a risk measure $\mathcal{R}$ is **coherent** if it satisfies the following properties:

1. **Subadditivity:** $\mathcal{R}(\xi + \eta) \le \mathcal{R}(\xi) + \mathcal{R}(\eta)$ for any random variables ξ and η.
2. **Positive homogeneity:** $\mathcal{R}(\lambda \xi) = \lambda \mathcal{R}(\xi)$ for all $\lambda \ge 0$.
3. **Monotonicity:** If $\xi \le \eta$ almost surely, then $\mathcal{R}(\xi) \le \mathcal{R}(\eta)$.
4. **Translation invariance:** $\mathcal{R}(\xi + c) = \mathcal{R}(\xi) + c$ for all constants $c \in \mathbb{R}$.

The expected value and the worst-case payoff are coherent risk measures, while the value at risk or the Markowitz risk measure are not coherent. Another advantage of conditional value at risk is that it is a coherent risk measure contrary to value at risk.
A risk measure is convex if it satisfies the 3 first properties of a coherent risk measure, in addition the 2 first properties already implies convexity.

9.3.1 Worst-Case Characterization of Coherent Risk Measures

A risk measure $\mathcal{R}$ is coherent if and only if there exists a set of probability measures $\mathcal{M}$ such that the risk measure can be written as the worst expected payoff over all probability distributions in this set:

$$\mathcal{R}(\xi) = \max_{q \in \mathcal{M}} \mathbb{E}_q[\xi] = \max_{q \in \mathcal{M}} \sum_{\omega \in \Omega} q_\omega \xi(\omega) \tag{9.6}$$

The probability distribution $\bar{q}$ maximizing this expression is called the **risk-adjusted probability measure**.
This interpretation is important because it shows that coherent risk measures behave as if agents evaluated uncertain outcomes under a pessimistic probability distribution that gives more weight to unfavorable scenarios.
For the particular case of CVaR, the set of admissible probability measures is:

$$\mathcal{M} = \left\{ q : q_\omega \le \frac{P_\omega}{\alpha}, \sum_{\omega \in \Omega} q_\omega = 1, q_\omega \ge 0 \right\} \tag{9.7}$$

The idea is that probabilities of bad outcomes can be amplified by a factor $1/\alpha$. Therefore, CVaR emphasizes the tail of the distribution and focuses on severe losses.
Another important result is that the maximizing probability measure $\bar{q}$ is a subgradient of the risk measure. Therefore, derivatives of a risk-adjusted payoff can be computed as:

$$\frac{\partial R(\xi)}{\partial a} = \mathbb{E}_{\bar{q}} \left[\frac{\partial \xi}{\partial a} \right] \tag{9.8}$$

This means that marginal decisions are evaluated using the risk-adjusted probabilities instead of the physical probabilities.

9.3.2 Back-Propagation

Definition 9.8. Back-propagation is the process by which forward prices are determined by the expected distribution of real-time prices, rather than the other way around.

Back-propagation highlights the fundamental role of balancing and real-time markets in electricity market design. Even though most electricity is traded in the day-ahead market, prices are ultimately anchored by the expected conditions of the real-time system.

9.3.3 Day-Ahead Market as a Forward Market

In the United States, the day-ahead market is interpreted as a financial forward market, while the real-time market is the physical spot market. Financial participants are allowed to trade in the day-ahead market even without owning physical assets.
The day-ahead market solves:

$$\begin{aligned} \min_{p_g} \quad & \sum_g MC_g p_g \\ \text{s.t.} \quad & \sum_g p_g = D^{DA} \qquad [\lambda^{DA}] \\ & 0 \leq p_g \leq P_g \end{aligned} \tag{9.9}$$

The real-time market solves:

$$\begin{aligned} \min_{p_g} \quad & \sum_g MC_g p_g \\ \text{s.t.} \quad & \sum_g p_g = D^{RT} \qquad [\lambda^{RT}] \\ & 0 \leq p_g \leq P_g \end{aligned} \tag{9.10}$$

The payoff of a market participant is: $\lambda^{DA} p^{DA} + \lambda^{RT}(p^{RT} - p^{DA})$.
For a conventional producer with $p^{DA} = p^{RT}$, the payoff becomes: $\lambda^{DA} p^{DA}$.
For a purely financial participant with $p^{RT} = 0$, the payoff becomes: $(\lambda^{DA} - \lambda^{RT}) p^{DA}$.
Hence, financial participants profit from differences between day-ahead and real-time prices and therefore contribute to price convergence between the two markets. In Europe, the design is different. The day-ahead market is usually considered as the physical spot market, while balancing markets are used to restore feasibility in real time. However, the economic logic remains equivalent, since balancing prices still reflect real-time system conditions.

Ressources adequacy

When we look at the planning problem at the long term, it can be interesting to think about the technology mix and especially the investment that we can do in the different technologies. This is what we call the capacity expansion planning problem.

10.1 Capacity expansion planning

The capacity expansion planning problem is a long-term planning problem that aims to determine which is the best technology mix with respect to the demand, the operating costs and the investment costs. It determines the best mix of technologies by minimizing the total cost of expanding and operating the power system over a long-term horizon. In its simplest form, the capacity expansion planning problem can be formulated as a two stage linear program where the first stage is the investment decision and the second stage is the operation decision. But this ignores the demand response. Beware that the demand is the consumption at a given price, but the load is the amount of power consumed if the price is zero.

10.2 Screening curve

To determine which is the best technology mix, we can solve this graphically with what is called a **screening curve**. The screening curve is a graphical representation of the total cost of operating a technology as a function of its capacity factor. For example here is the screening curve for these four technologies:

Technology	Investment cost [€/MWh]	Operating cost [€/MWh]
CCGT	16	25
OCGT	5	80
Nuclear	32	6.5
Biomass	2	160

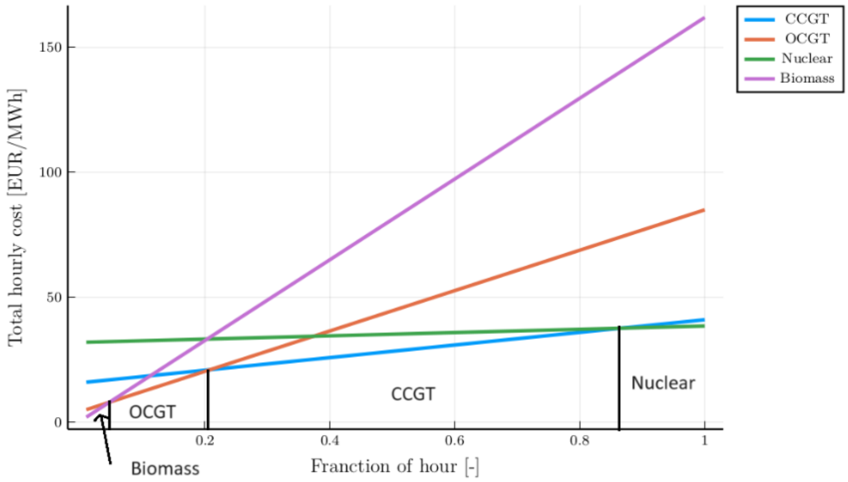

Figure 10.1: Screening curve for the four technologies

From the screening curve, we can determine the optimal technology mix by looking at the intersection points of the curves. For example, if the demand is such that the capacity factor is around 0, then the optimal technology mix is biomass. If the capacity factor is around 0.6 then the optimal technology mix is CCGT.
The investment costs can be categorized into 3 types:

- **Overnight cost (€/kW):** the cost that needs to be paid upfront per kW of investment.
- **Annualized fixed cost (€/kWy):** the cost that needs to be paid per year per kW of investment.
- **Hourly fixed cost (€/MWh):** the cost that needs to be paid per hour per MWh of investment.

We can convert the overnight cost to annualized fixed cost by using the following formula if the investment is paid every year for T years at a discount rate r:

$$FC = \frac{r\,OC}{1 - \frac{1}{(1+r)^T}} \tag{10.1}$$

If the investment is paid continuously for T years at a discount rate r, the formula is as follows:

$$FC = \frac{r\,OC}{1 - e^{-rT}} \tag{10.2}$$

The hourly fixed cost can be converted to annualized fixed cost by multiplying it by the number of hours in a year divided by 1000, which is 8.76.

10.2.1 Demand Response

The previous capacity expansion planning model assumes that demand is fixed and inelastic. However, in reality, consumers may adapt their consumption according to electricity prices. This is called **demand response**. It is important to distinguish:

- **Load**: the amount of electricity that would be consumed if electricity were free.
- **Demand**: the actual consumption at a given electricity price.

Therefore, load is always greater or equal to the demand. In this model, demand response is represented using block bids: (V_{lj}, D_{lj}) where:

- V_{lj} is the willingness to pay of load block l during slice j,
- D_{lj} is the maximum quantity that can be consumed.

The centralized capacity expansion problem with demand response becomes:

$$\begin{aligned}
\max \; & \sum_{j=1}^{m} \Delta T_j \left(\sum_{l \in L} V_{lj} d_{lj} - \sum_{g \in G} C_g p_{gj} \right) - \sum_{g \in G} I_g x_g \\
\text{s.t.} \; & \sum_{l \in L} d_{lj} - \sum_{g \in G} p_{gj} = 0 && [\rho_j] \\
& p_{gj} \leq x_g && [\mu_{gj}] \\
& d_{lj} \leq D_{lj} && [\nu_{lj}] \\
& p, x, d \geq 0
\end{aligned} \tag{10.3}$$

With ΔT_j the duration of slice j, D_{lj} the load during slice j, I_g the investment cost for generator g, C_g the variable operating cost for generator g, p_{gj} the production of generator g during slice j, d_{lj} the consumption of load block l during slice j, x_g the investment in technology g and V_{lj} the willingness to pay of load block l during slice j. The KKT conditions of this problem are:

$$\begin{aligned}
\rho_j free \perp \sum_{l \in L} d_{lj} - \sum_{g \in G} p_{gj} = 0 \\
0 \leq \mu_{gj} \perp p_{gj} - x_g \leq 0 \\
0 \leq \nu_{lj} \perp D_{lj} - d_{lj} \leq 0 \\
0 \leq p_{gj} \perp \Delta T_j C_g - \rho_j + \mu_{gj} \geq 0 \\
0 \leq d_{lj} \perp \rho_j - \Delta T_j V_{lj} + \nu_{lj} \geq 0 \\
0 \leq x_g \perp I_g - \sum_j \mu_{gj} \geq 0
\end{aligned} \tag{10.4}$$

The KKT conditions provide three important economic criteria:

- **Consumption criterion:** consumers consume electricity only if their willingness to pay exceeds the electricity price: $V_{lj} \geq \frac{\rho_j}{\Delta T_j}$
- **Production criterion:** generators produce electricity only if their marginal cost is lower than the electricity price: $C_g \leq \frac{\rho_j}{\Delta T_j}$
- **Investment criterion:** investments occur only if scarcity rents are sufficient to recover fixed investment costs: $I_g \leq \sum_j \mu_{gj}$

An important implication is that generation investments recover their fixed costs through **scarcity rents**, i.e. the difference between market revenues and variable operating costs.

10.3 Decentralized Capacity Expansion Planning

Instead of a centralized planner, investments can also emerge from a competitive electricity market. In an **energy-only market**, generators are remunerated only through energy sales and scarcity rents. There is no separate payment for installed capacity. The producer quantity adjustment optimization problem is:

$$
\begin{aligned}
\max \; & \sum_{j=1}^{m} (\rho_j - C_g)\Delta T_j p_{gj} - I_g x_g \\
\text{s.t. } & p_{gj} - x_g \leq 0 \qquad\qquad [\mu_{gj}] \\
& p_{gj} \geq 0
\end{aligned}
\tag{10.5}
$$

And the consumer quantity adjustment optimization problem is:

$$
\begin{aligned}
\max \; & \sum_{j=1}^{m} (V_{lj} - \rho_j)\Delta T_j d_{lj} \\
\text{s.t. } & d_{lj} - D_{lj} \leq 0 \qquad\qquad [\nu_{lj}] \\
& d_{lj} \geq 0
\end{aligned}
\tag{10.6}
$$

From the KKT conditions of these two problems, we can derive the scarcity rent earned by generator g in period j:

$$\mu_{gj} = (\rho_j - C_g)\Delta T_j \tag{10.7}$$

Therefore, investments are profitable only if total scarcity rents recover investment costs:

$$I_g \leq \sum_j \mu_{gj} \tag{10.8}$$

An important result is that the competitive equilibrium of the decentralized market satisfies the same KKT conditions as the centralized planner problem. Therefore, under perfect competition, energy-only markets lead to efficient investment and operation decisions. However, in practice, energy-only markets has also disadvantages. These are due to the low demand elasticity which leads to high price volatility, uncertain scarcity rents and high investment risk for some technologies.

10.4 Market design for resources adequacy

10.4.1 Value of Lost Load (VOLL) Pricing

The VOLL pricing are based on the idea of setting prices equal to the VOLL during periods of capacity shortage. This depends crucially on the accurate estimation of VOLL, which can be challenging. To model this problem, we introduce load shedding variables ls_j. The market model becomes:

$$
\begin{aligned}
VOLLP = \min \sum_{g\in G} I_g x_g + \sum_{j=1}^{m} \Delta T_j (VOLL \cdot ls_j + \sum_{g\in G} C_g p_{gj}) & \\
\text{s.t.} \sum_{g\in G} p_{gj} + ls_j = D_j \qquad\qquad & [\rho_j] \\
p_{gj} \le x_g & \\
p, x \ge 0 &
\end{aligned} \tag{10.9}
$$

This model replicates the outcome of ideal centralized planning process, as long as VOLL is correctly estimated but if VOLL is incorrectly estimated this model can be very inefficient.
From this model, we can derive two interesting KKT conditions. The first one is the market clearing condition:

$$\rho_j \perp D_j - ls_j - \sum_{g\in G} p_{gj} \tag{10.10}$$

And the second one is the price setting when shortage occurs:

$$0 \le ls_j \perp VOLL - \rho_j \ge 0 \tag{10.11}$$

This means:

- if load shedding occurs ($ls_j > 0$), then $\rho_j = VOLL$
- otherwise: $\rho_j \le VOLL$

10.4.2 Capacity Mechanisms

Because scarcity rents are uncertain and risky, many electricity markets introduce **capacity mechanisms**. The idea is to provide an additional payment for installing generation capacity. This mechanism can take various forms such as:

- Capacity auctions
- Reliability options
- Capacity payments
- Decentralized capacity mechanisms
- Strategic reserves

The main objective of capacity mechanisms is determining the targeted installed capacity. Consider the investment condition in (VOLLP) for the peaker technology g^*:

$$0 \le x_{g^*} \perp I_{g^*} - \sum_{j=1}^{m} \mu_{g^* j} \ge 0 \tag{10.12}$$

The peaker technology is only remunerated in periods of load shedding (for which $ls_j > 0$), in which case we have $\rho_j = VOLL$. Therefore,

$$\begin{aligned} I_{g^*} &\approx \sum_{j=1}^{m} \mathbb{1}[ls_j > 0] \cdot \mu_{g^*j} = \sum_{j=1}^{m} \mathbb{1}[ls_j > 0] \cdot \Delta T_j (VOLL - MC_{g^*}) \\ &= (VOLL - MC_{g^*}) \cdot \sum_{j=1}^{m} \mathbb{1}[ls_j > 0] \cdot \Delta T_j \end{aligned} \tag{10.13}$$

This value is called the **Cost Of New Entry** ($CONE$), which is the minimum revenue required to make a new investment profitable and it can be expressed as:

$$CONE \approx VOLL \cdot LOLP \tag{10.14}$$

where $LOLP$ is the **Loss Of Load Probability**. From this, we can derive a new optimization problem that includes a capacity mechanism. The CRM is a model that introduces an additional capacity payment ρ to the usual energy prices λ_j. The capacity market model is:

$$\begin{aligned} \max \int_0^{xd} VC(v)dv - \sum_{g \in G} I_g x_g - \sum_{j=1}^{m} \Delta T_j \left(VOLL \cdot ls_j + \sum_{g \in G} C_g p_{gj} \right) & \\ \text{s.t. } xd = \sum_{g \in G} x_g \qquad\qquad\qquad\qquad & [\rho] \\ \sum_{g \in G} p_{gj} + ls_j = D_j \qquad\qquad\qquad\qquad & [\lambda_j] \\ p_{gj} \le x_g \qquad\qquad\qquad\qquad & \\ p, x \ge 0 \qquad\qquad\qquad\qquad & \end{aligned} \tag{10.15}$$

Hence, generators receive:

- energy revenues through λ_j,
- capacity revenues through ρ.

Capacity mechanisms therefore reduce investment risk and improve long-term resource adequacy.